

AND-rPPG: A novel denoising-rPPG network for improving remote heart rate estimation

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ABSTRACT

Heart rate (HR) estimation is an essential physiological parameter in the field of biomedical imaging. Remote Photoplethysmography (r-PPG) is a pathbreaking development in this field wherein the PPG signal is extracted from non-contact face videos. In the COVID-19 pandemic, rPPG plays a vital role for doctors and patients to perform telehealthcare. Existing rPPG methods provide incorrect HR estimation when face video contains facial deformations induced by facial expression. These methods process the entire face and utilize the same knowledge to mitigate different noises. It limits the performance of these methods because different facial expressions induce different noise characteristics depending on the facial region. Another limitation is that these methods neglect the facial expression for denoising even though it is the prominent noise source in temporal signals. These issues are mitigated in this paper by proposing a novel HR estimation method *AND-rPPG*, that is, A Novel Denoising-rPPG. We initiate the utilization of Action Units (AUs) for denoising temporal signals. Our denoising network models the temporal signals better than sequential architectures and mitigate the AUs-based (or face expression-based) noises effectively. The experiments performed on publicly available datasets reveal that our proposed method outperforms state-of-the-art HR estimation methods, and our denoising model can be easily integrated with existing methods to improve their HR estimation.

1. Introduction

The COVID-19 pandemic poses a massive challenge for the existing world's healthcare system. It necessitates a significant transformation to telehealthcare, which utilizes digital and communication technologies to provide health care services remotely [1]. Currently, doctors usually prefer non-contact devices and applications to examine their patients and manage healthcare. In this direction, remote-PPG (rPPG) plays a pivotal role for doctors and patients to remotely analyze the heart and respiration rates using the camera sensors [2] and thereby perform cardiac disease monitoring and chronic diseases treatment therapies [3, 4]. Apart from telehealthcare, the rPPG is extensively explored in biometrics for liveness detection [5], spoof detection [6] and Deepfake detection [7]; and affective computing for micro-expression spotting [8], micro-expression recognition [9] and stress analysis [10]. Unfortunately, the applicabilities above offer limited efficacy because rPPG estimation can be spurious when the face video contains facial deformations, illumination variations, and camera sensor noise. Thus, we propose a heart rate (HR) estimation method in this paper that mitigates

these issues to improve the efficacy of rPPG.

The rPPG is estimated by observing the blood flow variation inside the carotid arteries of the human face, which results in heartbeats. The variations result in the subtle skin color change [11] and head motion [12] that are imperceptible to humans but can be analyzed using a camera. For instance, methods [12,13] employ RGB color and motion variations, respectively to compute the blood flow variations and thereby infer HR. The variations are referred to as temporal signals. Usually, the temporal signals are extracted from several facial regions, and the extracted signals are eventually consolidated for rPPG estimation. The temporal signals can be extracted by employing extensive domain-specific knowledge, which is usually PPG information and human skin models for rPPG estimation. For instance Ref. [14], used the green channel for the temporal signal extraction because it provides a better PPG signal than red and blue channels and [13] employs the optical skin characteristics for the extraction. In contrast, the temporal signal is obtained in method [12] by the motion variations of facial landmark points in the vertical direction because PPG information is prevalent in the vertical direction. Alternatively, the color and motion

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variations can also be analyzed using Deep learning architectures [15–18], for temporal signal extraction. Subsequently, the temporal signals from a face video are consolidated to extract the rPPG signal, and thereby HR is estimated using the rPPG signal.

In the constrained environment, both color and motion variations give accurate rPPG estimation to provide correct HR. Unfortunately, this HR estimation is incorrect and rPPG estimation is erroneous in real-world scenarios because the temporal signals are affected by the noise due to facial deformations and illumination variations [19]. Thus, several temporal signal denoising methods have been proposed in the literature to mitigate the noise. However, these denoising methods are ineffective in all possible scenarios. The major limiting factor of these denoising methods is that they utilize the same knowledge to denoise all the temporal signals derived from different facial regions. However, different facial regions contain different noise attributes. For instance, lip regions are mainly affected when smiling, whereas other regions like the forehead region are the least affected. Another limiting factor is that the facial expression is not explored in the literature to understand the noise characteristics even though the facial expression is the prominent noise source in temporal signal extraction. Moreover, the Deep Learning-based temporal signal denoising methods usually employ sequence models [20], but the sequence models are incapable of modeling the long sequences [21,22], which is the case with temporal signal modeling.

This paper proposes a novel HR estimation method, *AND-rPPG*, that is, *A Novel Denoising-rPPG*. It improves the HR estimation by denoising the temporal signals using action units (AUs) and Deep Learning architecture. In order to denoise the long sequence temporal signal, we propose a novel rPPG denoising method based on temporal convolution network (TCN) architecture. Since different facial regions contain different noise characteristics because they are affected differently by facial expressions, we utilize different networks to denoise temporal signals obtained from different facial regions. Along with the temporal signal, we provide AUs as the input to the networks to provide relevant information regarding the facial expressions. Our primary research contributions are:

1. To the best of our knowledge, we initiate the utilization of AUs for denoising the temporal signals. It is motivated by the fact that AUs provide the relevant information about the facial expressions, and the temporal signals are extensively affected by these expressions.
2. We propose utilizing different TCN networks to denoise different facial regions effectively because different regions are affected differently by different noises, and sequential networks are incapable of modeling long sequences.
3. The experimental results demonstrate that the proposed method outperforms state-of-the-art rPPG based methods on publicly available datasets. Moreover, it shows that our temporal signal denoising method can easily be integrated with existing methods to improve performance.

The paper organization is as follows. The next section provides a brief explanation of the related work for augmenting the understanding of the proposed method. Section 3 describes the proposed method. The subsequent section presents the experimental results. The conclusions are provided in the last section.

2. Related work

2.1. Conventional remote HR estimation

HR is the number of times a heart beats per minute. It can be computed using contact or non-contact techniques. Contact techniques require the proper touch between the sensor and human skin, like electrocardiography (ECG) or photo-plethysmography (PPG). Hence, the pulse signal is corrupted when the touch is disturbed due to

inevitable human movements, which thereby renders the contact techniques ineffective for unobtrusive HR monitoring of neonates, persons performing exercises, and skin-damaged patients [23]. Moreover, these techniques are incapable of performing telehealthcare, which is an essential requirement in the current COVID-19 pandemic situation to avoid the spread of infection. In contrast, the non-contact techniques avoid touch between human skin and sensor; thus, their applicability is proliferating. In this direction, HR can be computed remotely using the camera sensor, known as the remote HR estimation method.

The existing remote HR estimation methods mainly perform the following tasks: i) region of interest (ROI) extraction; ii) temporal signal estimation; iii) denoising extracted temporal signals; iv) consolidating the denoised signal for rPPG estimation; and v) eventually, HR estimation using rPPG. There exist several ways to perform these tasks in the literature, and the choice of ways to accomplish these tasks determines the efficacy of the HR estimation. The conventional HR estimation methods usually incorporate domain knowledge to accomplish the tasks. For instance, method [24] performs the temporal signal extraction using RGB color variations of the face regions followed by independent component (ICA) based blind source separation (BSS) for rPPG extraction. While temporal signals are extracted by tracking facial landmark points and rPPG is estimated using principal component analysis (PCA) based BSS in method [12]. Similarly, green color variations are employed in method [19] for temporal signal extraction, followed by Kurtosis-based BSS for rPPG estimation. These methods are ineffective when the face video contains head movement, facial deformations, and illumination variations.

2.2. Deep learning-based remote HR estimation

Currently, technological advancement results in applying Deep Learning as a de facto tool for solving real-world problems [25]. The remote HR estimation is no different in this aspect. Deep Learning can be applied to remote HR estimation in either a non-end-to-end manner or an end-to-end manner wherein Deep Learning architectures are used to accomplish some or all the HR estimation tasks, respectively.

Several well-known non-end-to-end methods [26–29] first extract rPPG signals using handcrafted feature extraction and subsequently employ Deep learning architectures for HR estimation. In Ref. [26], the rPPG signal extracted from CHROM [13] is transformed to time-frequency representation and then provided to VGG15 for HR estimation. Similarly [27–29], extracts the Spatio-temporal feature maps and feeds them to the network. The maps are obtained in Ref. [27] using spatial decomposition and temporal filtering while [28,29] obtained the map by consolidating the information from several face ROIs. These methods offer limited performance because they require stringent preprocessing and avoid global facial information [11]. In contrast, the non-end-to-end method DeeprPPG [30] employs spatiotemporal convolutions to compute the rPPG signals and subsequently consolidate the signals. Likewise, ETA-rPPGNET [31] divides the input video into segments and applies a time-domain segment subnet to extract the spatial and temporal information. These methods are not generalized due to head movement, facial deformation, and exposure [32].

Similarly, end-to-end Deep learning methods have been proposed in the literature, which takes face videos for HR estimation. For instance, HR-CNN [15] first extract the rPPG signal from face videos using a 2D CNN network and subsequently feed the signal to 1D CNN for HR estimation. Similarly, Deep-Phys [16] feeds the normalized difference frames to the convolutional attention networks for rPPG estimation. The rPPG estimation requires proper analysis of both spatial and temporal variations; thus, PhysNet [17] utilizes Spatio-temporal networks for rPPG estimation. Likewise, AutoHR [18] uses novel Temporal Difference Convolution (TDC) for motion estimation and feed it to neural architecture search, [20].

Both non-end-to-end and end-to-end Deep learning-based HR estimation methods require appropriate loss functions selection for training.

In the realm of rPPG, the loss functions can be defined using negative Pearson loss [17] which constraints the signal variations in the time domain or signal-to-noise ratio (SNR) [15] loss which imposes spectral peak localization constraint in the frequency domain. Kindly note that the efficacy of Deep learning for rPPG estimation is limited because these methods utilize the same knowledge to denoise all the temporal signals derived from different facial regions. However, different face regions are differently affected by different noise-inducing mechanisms like facial expressions.

2.3. Denoising rPPG signal

In conventional and Deep Learning-based HR estimation methods, the rPPG estimation can be erroneous when the temporal signals are affected by facial expressions and illumination variations. Thus, it is advisable to: i) find the facial regions that are least affected and utilize their temporal signals [33]; ii) remove the signals derived from the regions that contain facial movements [19]; or iii) rectify the temporal signals to mitigate these issues. For instance, the temporal signals are rectified in Ref. [34] using illumination rectification and non-rigid motion elimination. Likewise, illumination variations estimated by analyzing brightness are mitigated in Ref. [35]. Unfortunately, both these methods have limited applicability because these illumination models rely on the background conditions [36]. Similarly [37], estimates the facial motion by tracking the facial boundaries and mitigate it from the pulse signal. The estimation is error-prone because the tracking is susceptible to facial poses. The rectification can also be performed by: analyzing the consistency between ROIs using the self-adaptive matrix completion [38]; mitigating the non-stationary trends by employing Detrending filter [39]; leveraging the periodic properties of pulse signals using pulse modeling [11]; and reducing the noise due to head movements and illumination using inverse attention map of face, hair and background regions [20].

These denoising methods are ineffective because they neglect the facial expressions, which is a crucial noise-inducing source. Another limiting factor is that these methods utilize the same information to denoise the temporal signals computed from different facial regions instead of considering different facial regions to remove the noise. Another problem with Deep Learning-based denoising methods is that they consider sequential deep learning architectures to model the temporal signal, and it is insufficient to model the long sequences [21].

3. Proposed method

The proposed remote HR estimation method, **AND-rPPG** is presented in this section. It first divides the input video into several non-overlapping clips of fixed size, and HR is estimated corresponding to each clip. In this direction, the entire face present in the clip is detected and divided into different regions. Subsequently, the temporal signal for each region is extracted and denoised using region-specific Deep Learning networks. To this end, we propose a novel network known as Denoising-TCN, which is based on TCN architecture and requires temporal signal and AUs. The TCN-based is considered because it is capable of modeling long sequences more effectively than sequence models [21]. Moreover, the AUs are provided to the Denoising-TCN to provide the relevant information about the facial expression. In essence, each Denoising-TCN automatically learns how to denoise the temporal signal extracted from a specific facial region and corrupted by facial expressions. Eventually, the denoised temporal signals are used to estimate the pulse signal and compute the HR. The flow diagram of our proposed method can be visualized from Fig. 1.

3.1. Extracting ROI

Pulse signal information is present in facial skin, and it needs to be extracted from the input clip. In this direction, we find the discriminatory facial points called landmark points that delineate the face boundary and its subparts like eyes, mouth, nose, and eyebrow. We employ the CLNF Openface [40] to extract 68 landmark points from the first frame of the face. These landmark points are utilized to extract the different ROIs. We aim to divide the face into regions so that each denoised network learns to mitigate the noise induced by facial expressions in a specific facial region. It is because facial expressions affect different face regions differently. To achieve the aim, we ROIs are given by dividing the face region into the Delaunay triangular regions using [41]. In our case, a Delaunay triangulation refers to dividing the face region into triangles using landmark points as vertices. A Delaunay triangles is formed using three landmark points such that the triangle's circumcircle does not contain any other landmark point. Kindly note that we avoid those ROIs that contain the eyes and eyebrow regions because they are easily affected by unavoidable eye-blinking. To identify the location of these ROIs, we used landmark points of regions like eyes, lips, and nose. These regions are extracted by the convex hull of facial landmark points corresponding to these regions [8]. Moreover, it is

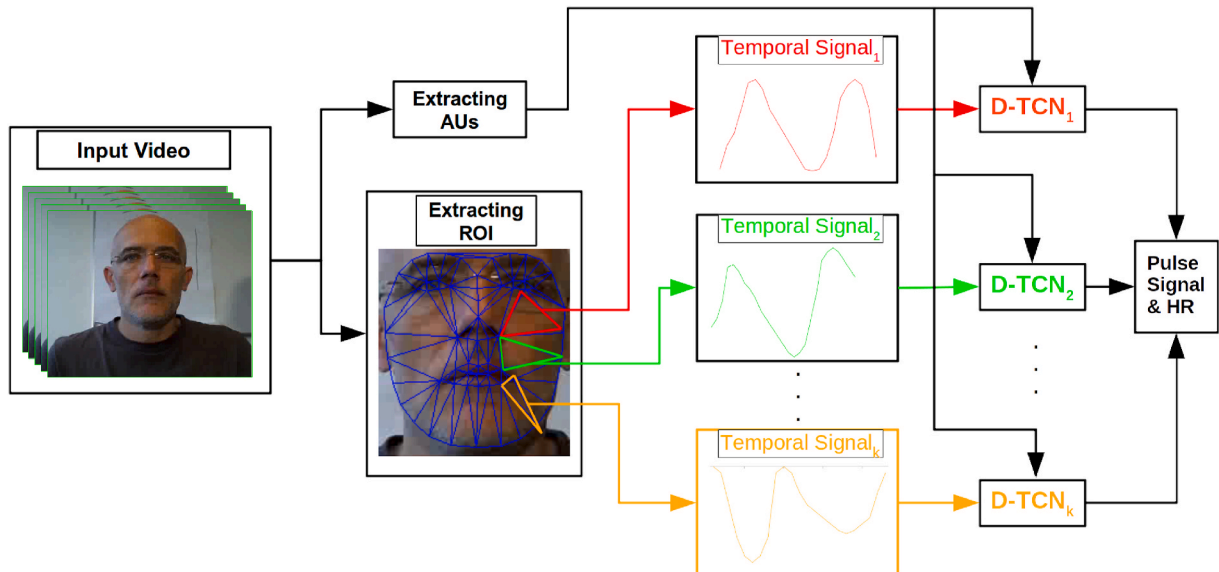


Fig. 1. Flow graph of the proposed method, **AND-rPPG**.

observed in Ref. [19] that the facial boundary pixels are easily affected by facial deformations and better performance can be expected by removing them. Hence, we estimate the boundary pixels by joining the landmark points belonging to the facial boundary. Moreover, we remove these pixels by utilizing the morphological erosion operation [42]. The erosion is performed by setting the filter size of 5 pixels. Moreover, we do not consider the Delaunay triangles belonging to lip regions and those triangles lying between the nose and lip regions because these regions are usually affected by facial deformations and contain small regions to provide relevant information. Thus, the total number of ROIs extracted from the face in our case is 36.

3.2. Extracting AUs

The facial deformations induced by facial expression can introduce noise in the extracted temporal signals. Our method aims to mitigate such noise from the temporal signal of each ROI; thus, it requires the proper evaluation and analysis of facial expressions for denoising. AUs are frequently employed in the literature to analyze facial expressions based on their facial appearance [9]. Thus, we utilize AUs in our proposed method to analyze relevant information regarding facial deformation induced by expressions. We employ the extensively utilized CLNF Openface [40] to extract the AUs from the face. CLNF-Openface is trained on the DISFA dataset [43] using the Deep-CNN model consisting of five convolutional layers, two fully-connected layers, and a linear layer. It takes an input clip and extracts the 18 AUs from each frame. Each extracted AU can detect: i) the presence/absence of a particular facial deformation in terms of 0 and 1; or ii) the probability of the existence of a particular facial deformation. We have only utilized the probability of the existence of AUs in our proposed method.

3.3. Temporal signal extraction

In this subsection, a temporal signal is extracted corresponding to each ROI by analyzing the color intensity variations. We preferred green channel intensity variations over red and blue channel intensity variations because the red channel provides the strongest PPG signal [14]. The temporal signal is the average green channel intensity variations of the pixels belonging to the Delaunay triangle. Mathematically, the temporal signal, T_j corresponding to j th Delaunay region is given by:

$$T_j = \frac{\sum_i p_g^i(1)}{L(1)_j}, \frac{\sum_i p_g^i(2)}{L(2)_j}, \dots, \frac{\sum_i p_g^i(n)}{L(n)_j} \quad (1)$$

where $\sum_i p_g^i(i)$ represents the sum of green intensities of the pixels belonging to j th Delaunay region in the i th frame, $L(i)_j$ represents the total number of pixels in the j th Delaunay region in the i th frame, and n is the total number of frames in an input clip.

The extracted temporal signal contains the noise generated by facial movement, respiration rate, and illumination. We mitigate several noises, including respiration rate noise, by applying fourth-order Butterworth bandpass filtering on the temporal signals [44]. Typically, the human heart beats at the rate of 42–240 beats per minute (BPM) [19]. Thus, the range of the bandpass filtering is set within a range of 0.7 (or 42/60) to 4 (or 240/60) Hz. Likewise, noise due to illumination variations introduces non-stationary trends in the temporal signals. It is mitigated by applying the Detrending filter [11].

3.4. Denoising TCN

The temporal signals extracted from different facial regions can be spurious due to the facial expressions. Hence, the signals are denoised in this subsection to improve HR estimation. To this end, we propose a novel Denoising-TCN model. As discussed, the different noise-inducing factors result in different noise characteristics depending on facial regions. Thus, we apply several Denoising-TCN models, such that each

model denoises the temporal signal extracted from a particular facial region. It enables the Denoising-TCN corresponding to an ROI to learn how to mitigate the noise generated by facial expressions in the temporal signal of the considered ROI. Since a significant noise-inducing source in the extracted temporal signal is facial expression, our model analyzes the AUs (which provide information about facial expressions) to denoise the temporal signal. In essence, each Denoising-TCN model denoises only a temporal signal extracted from a particular ROI by utilizing the AUs of the entire face. Moreover, our proposed model employs TCN-based architecture to model the long sequence of temporal signals because it is shown to be more effective for modeling long sequences than extensively explored sequential models [21].

3.4.1. Our Denoising-TCN network architecture

The network architecture of Denoising-TCN is discussed in this subsection. It is also depicted in Fig. 2 for visualization. It requires the temporal signal and the AUs to learn the facial expressions, and its output is the corresponding denoised temporal signal. The input dimension of Denoising-TCN is $19 \times f$, where f denotes the number of frames, and 19 is the concatenation of 18 AUs and one temporal signal. The first block of Denoising-TCN performs 2D convolution using seven filters of size 19×7 with the same padding in the temporal direction and no padding in the other direction (refer to Subsection 4.3 for this parameter selection). Therefore, the resulting output size is $7 \times 1 \times f$. The resultant is then passed to the average pooling layer, applied in just one direction to reduce the dimension and make it the same size as temporal signals, which is $1 \times f$. It is followed by adding a Rectified Linear Unit (ReLU) activation layer. We used the ReLU layer to introduce non-linearity because it is more computationally efficient than other activation layers [45]. Subsequently, we pass the resultant from two sequential non-causal TCN (ncTCN) blocks. The convolutions can be applied in TCN using either causal or non-causal ways. The causal convolution is computed over each time step of the sequence where a time step cannot depend on the activation computed from the future time step sequences. In contrast, the non-causal operations depend on the future time steps. Since we know the complete temporal signal, we apply non-causal convolution to consider past and future time steps. For both the ncTCN blocks in Denoising-TCN, the input size is $1 \times f$, on which 1D convolutions are applied using the seven filters of size 1×7 with the same padding. The resulting output is of size $7 \times 1 \times f$. Then, we apply average pooling in one direction to reduce the dimension and make it the same size as the temporal signal. Finally, we apply a ReLU activation. The final output of Denoising-TCN is of size $1 \times f$, which is a denoised temporal signal. Furthermore, the performance is improved by adding the skip connection (inspired from Ref. [46]) in both the ncTCN blocks. As shown in Fig. 2, it is added between the input and the second last output layers of each ncTCN block. Also, we increase the receptive field by setting the dilation rate to 2 and 4 for the second and third blocks of Denoising-TCN, respectively.

In this way, we design different Denoising-TCN networks corresponding to each ROI. Kindly note that we also utilize dropout to improve performance. The remaining parameter details are provided in Section 4.2.

3.4.2. Data augmentation

The existing rPPG datasets are unbalanced as most of these dataset samples contain the HR in the range of 70–90 BPM. However, the normal HR range is 40–240 BPM. If our Denoising-TCN is trained using the unbalanced datasets, the trained network will be biased to the samples containing HR between 70 and 90 BPM. This issue is mitigated by introducing new samples in the range of 40–70 BPM and 90–240 BPM using the linear interpolation data augmentation technique. We have applied the downsampling and upsampling techniques to the temporal signals. Alternatively, it is possible to apply these techniques to the video frames. Both these methods work similarly and provide similar performance. However, we need to perform time-expensive temporal signal

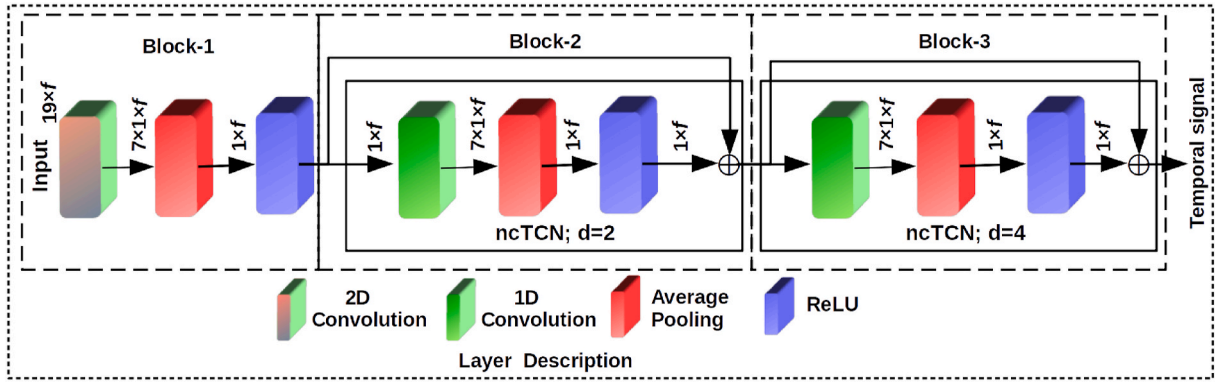


Fig. 2. The architecture of our proposed Denoising-TCN. It takes the AUs and temporal signal extracted from an ROI and provides the Denoised temporal signal. Input is passed to the 2D convolution, followed by average pooling and ReLU. The Middle and last blocks are ncTCN blocks that require the same input and output size.

extraction and frame interpolations when data augmentation is applied to video frames. In contrast, these time-expensive steps are avoided when data augmentation is performed on temporal signals. Therefore, we apply data augmentation to the temporal signals. To this end, we down sample the temporal signals with a factor of 2 and 3 to generate the samples having double and triple HR, respectively. Similarly, we up sample the temporal signals to generate lower HR samples. The upsampling and downsampling of the temporal signals can generate few samples outside the valid HR range of 40–240 BPM. Kindly note that we do not consider such samples for training our Denoising-TCN.

3.5. Pulse signal extraction and HR estimation

In this subsection, the denoised temporal signals from all the facial regions are consolidated to compute the rPPG signal and estimate the HR. Several BSS techniques and periodicity measures have been extensively explored in the literature for such consolidation. Amongst these works, we utilize the Multi-kurtosis optimization [19] that utilizes both BSS and periodicity simultaneously to provide rPPG estimation. Subsequently, we apply the Fast Fourier transform (FFT) on the pulse signal to compute the pulse spectrum for computing the HR. The number of samples in FFT depends on the frame rate of the clip. The HR is given by the frequency containing the maximum amplitude in the pulse spectrum. Mathematically, HR, h is given by:

$$h = f_r \times 60 \quad \text{where} \quad f_r = \underset{f}{\operatorname{argmax}} P[f] \quad (2)$$

where $P[k]$ denotes the amplitude of pulse spectrum at frequency k and f_r is the frequency containing maximum amplitude in the pulse spectrum. The frequency is multiplied by 60 to compute the HR because HR is given by the number of heartbeats in 60 s.

4. Experimental results

4.1. Dataset and metrics

We used the publicly available datasets, UBFC-rPPG [47] and COHFACE [48] to perform the comparative analysis of our proposed method, **AND-rPPG**. The COHFACE dataset contains 160 face videos and physiological information, acquired from 40 healthy subjects in realistic conditions. The dataset is split into training and testing sets consisting of 24 (60%) and 16 (40%) subjects, respectively. Likewise, the UBFC-rPPG dataset contains 42 videos acquired from 42 subjects. Its training and testing sets consist of 28 (67%) and 14 (33%) subjects, respectively. We follow these training and testing split settings in our proposed method for comparative analyses. The duration and frame rate of the videos for the COHFACE dataset are 1 min and 20 fps,

respectively. Similarly, the duration and frame rate of the UBFC-rPPG dataset videos are about 2 min and 30 fps, respectively. The ground truth is provided along with the datasets. It is acquired as a pulse signal using a pulse oximeter during the dataset creation. For performance evaluation, we use the standard deviation (SD), the mean absolute error (MAE), the root mean square error (RMSE), and the Pearson's correlation coefficient (r) between the ground truth HR and estimated HR.

4.2. Implementation details

The proposed method is implemented in python using the Pytorch library. The experiments are conducted on the NVIDIA V100 GPU server and Intel Xeon Gold 6132 processor with 192 GB of RAM. Kindly note that the proposed method requires training of several Denoising-TCN models, each corresponding to an ROI. The training is performed by maximizing the SNR loss function [15] using Adam optimizer. To this end, the learning rate and the maximum number of epochs are set to 0.001 and 500, respectively. However, we stop the training if validation loss does not decrease in 20 epochs. Also, the momentum, dropout, and weight decay are set to 0.9, 0.1 and 0.000 1, respectively.

4.3. Parameter selection

The first step in the HR estimation is dividing the input video into several clips of equal size. The size of the clip is set as 4 s, and the corresponding signal acquired from the pulse-oximeter is used as the ground truth. This size plays a crucial role. If it is set to a small value, it is challenging to overcome the loss of information [31]. In contrast, if the size is set to a large value, then fewer training samples are available for training the model, and significant information, like HR variability, cannot be inferred. It is shown in Refs. [27,31] that 4 s is a considerable size for clip extraction. Thus, we consider the clip size of 4 s in our proposed method.

Another parameter of our proposed method is the size of the filter used in Denoising-TCN. The filter size depends on the size of the input. Its first dimension is of size 19, which is equal to the size of the first dimension of input, and its second dimension is our hyperparameter (k). We found out that when we took 19 filters of size seven, our Denoising-TCN performed best. If we change the size of the second dimension of the filter (hyperparameter k), then the performance of the proposed method varies, which can be visualized from Fig. 3. This figure depicts that the performance of the proposed method is best when the value of k is 7 for both datasets.

4.4. Comparative evaluation

In this section, we compare our proposed method with state-of-the-

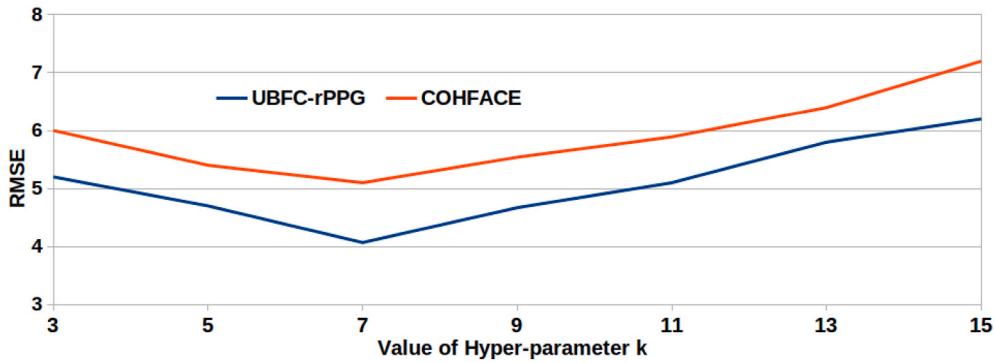


Fig. 3. The RMSE of our proposed method *ANR-rPPG* varies with filter size (hyperparameter value k). The best performance is achieved when k is equal to 7.

art methods on COHFACE and UBFC-rPPG datasets, and the results are shown in Tables 1 and 2. Kindly note that we reimplemented several state-of-the-art methods for fair comparative analysis. In Table 1, we show comparative performance for 20-s, whereas, in Table 2, we replace their ROIs with our Delaunay triangular region with and without Denoising-TCN. We used the publicly available codes¹ and their pre-defined parameters setting to evaluate these methods for comparative analysis. Also, we adhere to the training and test set distributions provided in Refs. [15,48] for fair comparative analysis.

Table 1 depicts that our proposed method outperforms the state-of-the-art methods. Our proposed method outperforms the 2SR [49], ICA [24], Chrominance-rPPG [13], and RAHR [19] because these methods are insufficient to denoise the noise-induced from facial expressions. In contrast, our proposed method explicitly analyzes the facial expression to denoise the temporal signals and improve the performance. Likewise, our method also surpasses the performance of AHRE [50], Fusion-EL [51], SHR [2], and MOMBAT [11] because these methods neglected the facial expressions, and they used the same knowledge for analyzing the relevant information in all the ROIs. However, the different sub-regions of the face have different expressions that induce different types of noise. Due to the same reason, our proposed method also outperforms the following Deep learning-based methods: Deepphys [16], PhysNet [17], HR-CNN [15]. Apart from these issues, another issue prevalent in Two Stream [52], and META-rPPG [53] that result in performance degradation is that these models used the LSTM architecture to model the rPPG signals. Unfortunately, these sequential architectures are insufficient to model the long sequence of temporal information [21]. Unlike these systems, our Denoising-TCNs mitigates the noise from the temporal signals by considering the facial location and analyzing the facial expressions using AUs. Thus, our systems perform better than these systems.

For a more thorough comparative performance evaluation of our proposed Delaunay triangular region and Denoising-TCN models, we reimplemented these state-of-the-art methods by adding our Delaunay triangular region and Denoising-TCN models. The proposed Denoising-TCN model denoises the temporal signals of different state-of-the-art methods. The corresponding results are shown in Table 2. Our Denoising-TCN can be easily integrated with existing methods that provide temporal signals. Thus, we cannot apply our Denoising-TCN on end-to-end Deep learning-based methods because the processing or working of each layer in an end-to-end Deep learning-based method is not known, and hence, the layer providing the temporal signals is not known. Thus, the performances of end-to-end Deep learning-based methods are not reported in the Table. It can be observed from the Table that there is a performance improvement in state-of-the-art methods when our Denoising-TCN model is incorporated. It happens because our added models mitigate the noise in temporal signals due to

facial expressions and thereby improve the pulse signal extraction.

4.5. Ablation study

This subsection provides the details of the experiments conducted to thoroughly inspect the importance of AUs, Denoising-TCN, and different Delaunay regions in the proposed method. These experiments are conducted by modifying the proposed method, and results are reported in Table 3.

The experiment *Without-AUs* is performed to understand the importance of AUs in Denoising-TCN. It is conducted by modifying the proposed method by removing AUs from Denoising-TCN. It can be observed from the table that our proposed method *AND-rPPG* outperforms *Without-AUs*. It indicates that the performance can be improved by analyzing AUs because it provides the relevant facial expression information, which can be mitigated from the temporal signals to improve HR estimation. Similarly, we analyze the importance of the Delaunay triangular region in *AND-rPPG* by conducting the *One-Region* experiment. The experiment is formed by considering the entire face as one region instead of dividing the face into different ROIs. The table shows that our proposed method outperforms *One-Region*, which agrees with the findings in Ref. [19]. Thus, it is advisable to consider several ROIs rather than one ROI considering the entire face for better HR estimation. The experiment *Raw-Temporal* is performed by avoiding the bandpass filter to the temporal signal of the proposed method. The table indicates that the proposed method outperformed the *Raw-Temporal*. It advocates the utilization of bandpass filtering to remove the noise from temporal signals before computing HR. This observation is also well-established in the literature [11,19].

We performed some additional experiments to understand the importance of Denoising-TCN. In these experiments, we replace the proposed method's Denoising-TCN with the sequence models like LSTM and RNN to form the *LSTM-exp* and *RNN-exp* experiments, respectively. The table indicates that better performance can be expected when our Denoising-TCN is utilized rather than the sequential models in the proposed method. It is because these sequential models are insufficient to model the long sequences of temporal information [21]. Furthermore, kindly consider Fig. 4 to visualize the importance of our Denoising-TCN. It depicts an example where performance improves by incorporating our denoising-TCN. We normalize the ranges of ground truth and the signals for better visualization. In the example, the facial expression introduces the deformation after 2 s. Thus, the extracted pulse signal is erroneous from 2 to 3.5 s (refer to the red color pulse signal of the first row). However, this noise is mitigated by our Denoising-TCN (refer to the green color pulse of the fourth row). Hence, the predicted pulse and PSD are similar to the ground truth in the fourth and last row and give correct results. In contrast, the PSD and pulse signals of ground truth, without denoising in the first row, and the first and second block output of Denoising-TCN in the second and third row are different.

¹ <https://github.com/lokendra7/rPPG-Publically>.

Table 1

Comparative Analysis of the proposed method with other state-of-the-art methods.

	UBFC-rPPG				COHFACE			
	SD*	MAE*	RMSE*	r	SD*	MAE*	RMSE*	r
2SR [49]	17.89	15.95	11.65	0.12	22.30	20.97	25.98	-0.33
ICA [24]	12.80	6.95	13.60	0.74	13.83	8.89	14.55	0.42
Chrominance-rPPG [13]	5.21	3.21	6.14	0.68	11.61	10.15	12.69	0.20
AHRE [50]	7.00	6.15	7.92	0.50	8.10	8.27	11.31	0.19
Fusion-EL [51]	6.02	5.08	7.42	0.57	7.98	8.97	10.84	0.26
RAHR [19]	4.30	3.49	4.35	0.69	8.73	7.91	9.45	0.47
MOMBAT [11]	4.62	4.04	5.17	0.84	7.92	7.01	8.24	0.61
PhysNet [17]	4.47	4.12	5.10	0.83	8.85	8.63	9.36	0.54
Two Stream [52]	5.19	4.81	5.70	0.60	8.93	8.29	10.14	0.37
META-rPPG [53]	8.00	6.54	9.11	0.46	11.52	9.31	12.27	0.19
Deepphys [16]	8.73	6.27	10.82	0.65	9.01	8.25	14.71	0.28
SHR [2]	4.61	3.59	5.28	0.82	8.51	6.95	8.52	0.57
HR-CNN [15]	5.21	4.90	5.89	0.64	9.46	8.92	11.88	0.16
AND-rPPG	4.01	3.15	4.75	0.92	7.83	6.81	8.06	0.63

* values are in BPM.

Table 2

Performance Evaluation of our proposed method and state-of-the-art methods.

	Denoising-TCN	UBRF-rPPG				COHFACE			
		SD*	MAE*	RMSE*	r	SD*	MAE*	RMSE*	r
2SR [49]	Not Used	17.25	15.85	19.74	0.03	22.18	20.98	25.84	-0.32
	Used	10.98	12.95	11.60	0.12	19.05	18.41	21.98	-0.18
ICA [24]	Not Used	4.50	3.70	4.61	0.67	10.63	7.80	12.45	0.26
	Used	6.42	4.95	7.45	0.84	8.78	8.10	12.65	0.23
Chrominance-rPPG [13]	Not Used	4.50	3.70	4.61	0.67	10.63	7.80	12.45	0.26
	Used	3.92	3.01	4.01	0.71	7.96	6.32	10.99	0.22
AHRE [50]	Not Used	4.95	4.20	5.78	0.61	6.38	5.72	11.52	0.31
	Used	4.82	4.03	5.12	0.64	7.50	6.31	10.84	0.26
Fusion-EL [51]	Not Used	4.20	3.71	4.52	0.73	8.09	7.14	9.43	0.57
	Used	4.17	3.54	4.25	0.78	6.98	5.88	8.02	0.60
RAHR [19]	Not Used	4.50	3.70	4.61	0.67	10.63	7.80	12.45	0.26
	Used	3.28	2.98	3.45	0.81	6.20	5.62	7.21	0.63
MOMBAT [11]	Not Used	3.38	3.50	4.01	0.85	6.14	5.89	7.92	0.62
	Used	3.24	2.74	3.59	0.89	5.38	4.50	6.01	0.68
Physnet [17]	Not Used	3.85	3.63	5.29	0.94	7.9	8.59	11.60	0.36
	Used	3.58	3.41	4.99	0.95	5.85	6.63	8.29	0.74
Two Stream [52]	NA [†]	4.32	3.94	5.02	0.67	8.69	8.09	9.96	0.40
META-rPPG [53]	NA [†]	4.50	3.70	4.61	0.67	10.63	7.80	12.45	0.26
Deepphys [16]	NA [†]	7.42	5.71	8.58	0.70	7.80	6.89	13.89	0.34
SHR [2]	NA [†]	3.76	2.85	4.92	0.95	6.28	5.92	7.95	0.63
HR-CNN [15]	NA [†]	4.15	3.82	4.92	0.71	9.23	8.10	10.78	0.29
AND-rPPG	NA [†]	3.21	2.67	4.07	0.96	4.53	3.82	5.10	0.79

* values are in BPM. [†] NA mean not applicable.**Table 3**

Ablation study of our proposed method.

	UBFC-rPPG				COHFACE			
	SD*	MAE*	RMSE*	r	SD*	MAE*	RMSE*	r
<i>AND-rPPG</i>	3.21	2.67	4.07	0.96	4.53	3.82	5.10	0.79
<i>Without-AUs</i>	8.99	8.28	10.13	0.70	11.92	11.45	12.97	0.59
<i>LSTM-exp</i>	7.82	7.02	8.98	0.71	10.35	9.20	10.92	0.62
<i>RNN-exp</i>	8.95	8.76	9.52	0.69	12.48	11.60	13.85	0.56
<i>One-Region</i>	11.02	10.36	12.10	0.54	13.05	12.62	14.78	0.49
<i>Raw – Temporal</i>	14.24	13.65	15.00	0.41	15.25	14.19	16.37	0.42

* values are in BPM.

5. Discussion

We evaluate the performance of our proposed method for different time duration clips, and the results are reported in Table 4. It indicates that as the time duration of clips increases to a particular time duration, the performance of our proposed method increases. After that, it shows the debilitating behavior when the time duration increases. After 7 s, the number of training samples for our proposed method is diminutive as

the clip's duration increases. The table also indicates that the performance decreases when small size clips are considered because it is difficult to overcome the loss of information for small size clips [31].

Unlike most existing remote HR estimation methods in the literature, the proposed method required the position-specific ROI for training the different Denoising-TCN. It enables the utilization of different Denoising-TCN networks to denoise the temporal signals derived from specific facial regions. It is our requirement because different facial

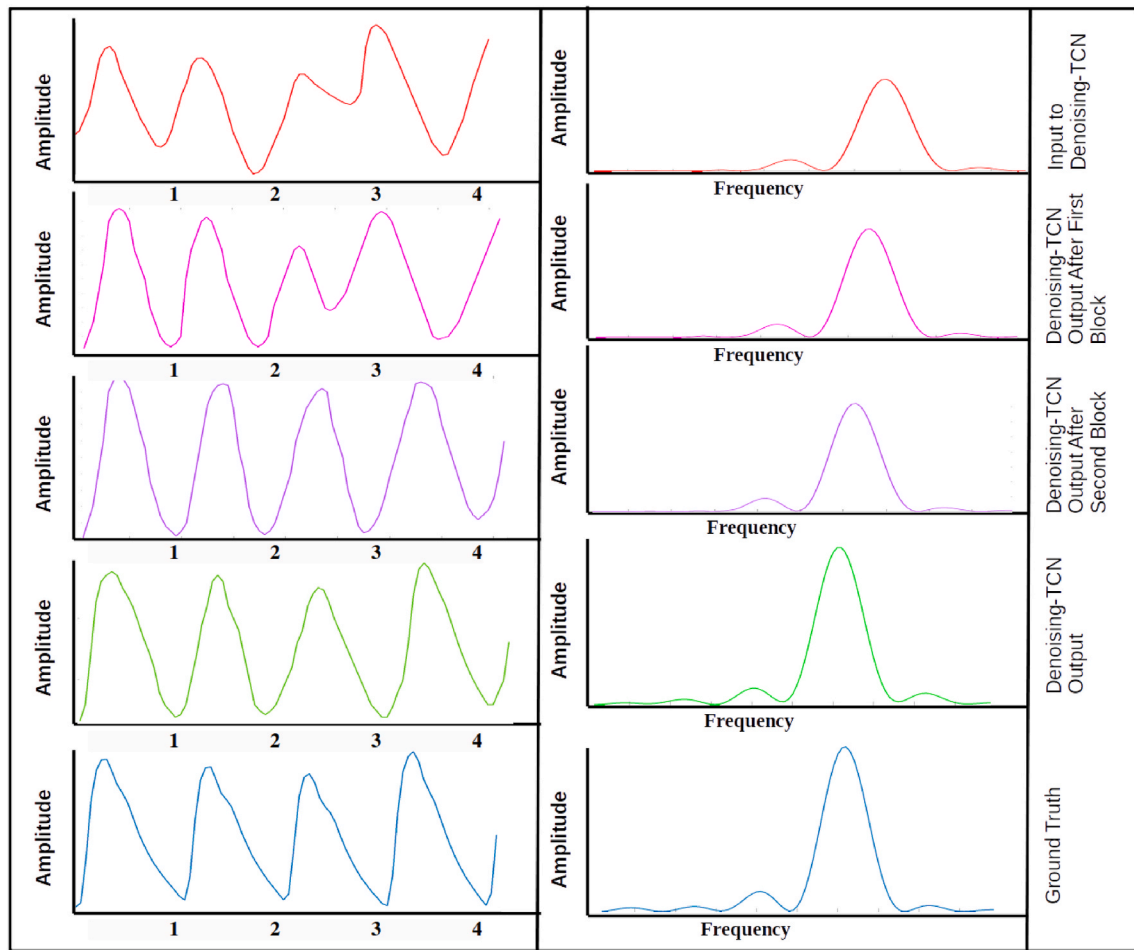


Fig. 4. Importance of Denoising-TCN. The left column denotes the pulse signal, and the middle column denotes the pulse spectrum density. The first row is input to the Denoising-TCN. The second, third, and fourth rows are the output of Denoising-TCN after the first, second, and final blocks, respectively. The last row indicates the ground truth of the corresponding sample.

Table 4

Performance evaluation our proposed method for different time duration clips.

	UBFC-rPPG				COHFACE			
	SD*	MAE*	RMSE*	r	SD*	MAE*	RMSE*	r
2-Second	4.95	3.61	6.46	0.90	7.04	6.45	8.20	0.74
3-Second	4.18	3.15	5.16	0.93	5.79	4.67	6.12	0.76
4-Second	3.21	2.67	4.07	0.96	4.53	3.82	5.10	0.79
5-Second	3.10	2.54	3.55	0.97	4.47	3.73	4.99	0.79
6-Second	3.00	2.17	3.12	0.98	4.03	3.12	4.70	0.80
7-Second	3.17	2.20	3.28	0.98	3.84	3.02	4.48	0.81
8-Second	3.29	2.23	3.55	0.97	3.38	3.57	4.95	0.79
9-Second	3.42	2.27	3.87	0.96	5.01	4.16	5.79	0.76
10-Second	3.54	2.32	4.05	0.96	6.85	5.64	7.16	0.71

* values are in BPM.

expressions induce different noise characteristics depending on facial regions. Our proposed method can be easily integrated with other state-of-the-art methods except for end-to-end Deep learning methods because we do not know what type of processing is performed by which layer and which layer is computing the temporal signals.

6. Conclusion

Remote HR estimation is an indispensable development in the field of affective computing and healthcare. Due to the COVID-19 pandemic, contact-based HR estimation is avoided, and it is the necessity of time to

use more accurate remote HR estimation methods. However, conventional and Deep learning-based rPPG methods perform poorly due to various noises induced in the pulse signal. Our novel denoising rPPG method has mitigated the facial expression-based noise from the temporal signal effectively. The facial expressions have not been explored in the literature to mitigate noise in the temporal signals. In this direction, we have incorporated AUs to analyze the facial expressions and proposed a novel denoising network, Denoising-TCN, to denoise the temporal signal extracted from specific ROI. The proposed method has utilized different networks to denoise specific ROI because different facial expressions induce different noise characteristics depending on

facial regions. The experimental results are performed on publicly available datasets, COHFACE and UBFC-rPPG. The results have revealed that our proposed method outperforms the state-of-the-art methods. Furthermore, the results have justified the utilization of Denoising-TCN and AUs for removing facial expressions-based noises from facial regions. Moreover, it has been shown that our Denoising-TCN can be easily integrated with other state-of-the-art methods to improve their performance. In the future, we will be looking to improve the HR estimation for the compressed dataset by utilizing Lagrangian techniques.

Declaration of competing interest

None Declared.

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